

HIGHER

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
3	(a)	(i)		Offspring/results are obtained quickly/ does not take a long time to get results/ short life cycle		1		1		1									
		(ii)		(Greater) confidence in result	1			1		1									
	(b)	(i)		either of the 2 flies with dark end to abdomen		1		1											
				Gametes correct (1) <table><tr><td>Gametes</td><td>R</td><td>R</td></tr><tr><td>r</td><td>Rr</td><td>Rr</td></tr><tr><td>r</td><td>Rr</td><td>Rr</td></tr></table> Mechanics correct (1)	Gametes	R	R	r	Rr	Rr	r	Rr	Rr		2		2		
Gametes	R	R																	
r	Rr	Rr																	
r	Rr	Rr																	
	(c)	(i)		All ratios 3: 1		1		1	1										
		(ii)		{Flies/ embryos/ they} died/ one or both flies were {sterile/ infertile} flies did not mate/ flies diseased/ eggs did not hatch/ develop Reject no offspring unqualified			1	1											
		(iii)		Results for red eyed flies and white eyed flies have been inserted in wrong columns/ OWWTE Do not accept that the wrong flies were mated because no cross could produce this ratio. Do not accept a mutation occurred			1			1 1									
		(iv)		<table><tr><td>Gametes</td><td>R</td><td>r</td></tr><tr><td>R</td><td>RR</td><td>Rr</td></tr><tr><td>r</td><td>Rr</td><td>rr</td></tr></table> Gametes correct 1 mark Mechanics correct 1 mark	Gametes	R	r	R	RR	Rr	r	Rr	rr		2		2		
Gametes	R	r																	
R	RR	Rr																	
r	Rr	rr																	
				Question 3 total	1	7	2	10	1	4									